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Anthropomorphism Effects on Social Robot Attitudes: Psychological Distance as a Mechanism and Product Information Type as a Boundary Condition

Completed Research Paper

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Abstract

This study examines how anthropomorphism influences consumer attitudes toward social robots by focusing on the cognitive mechanism of psychological distance and the moderating role of product information type. Using a 2×2 between-subjects online experiment with 206 participants, we manipulated anthropomorphism level (high vs. low) and product information type (attribute-focused vs. benefit-focused). The results show that anthropomorphized social robots are perceived as psychologically closer to consumers. Furthermore, psychological distance mediates the relationship between anthropomorphism and attitudes, but this effect occurs only when product information emphasizes concrete attributes. Under benefit-focused appeals, the indirect effect is not significant. These findings suggest that anthropomorphism does not universally enhance consumer evaluations, but that its persuasive effectiveness depends on the congruence between psychological distance and information framing, offering theoretical insights into human-robot interaction and practical implications for social robot marketing strategies.

Keywords: Social robot, anthropomorphism, psychological distance, construal level theory, product information type

Introduction

Social robots occupy an ambiguous position between technological products and social interaction partners. Unlike conventional technological products, social robots are designed to communicate with humans through social cues such as facial expressions, voice, gestures, and interactive responses (Breazeal, 2003; Fiore et al., 2013). These cues can lead consumers to interpret robots not merely as functional tools, but as social entities capable of interaction, intention, and relational engagement. As a result, consumer evaluations of social robots may depend not only on what functions they provide, but also on how consumers construe the robot's social presence and role.

This ambiguity creates an important challenge for the productization and communication of social robots. Firms and designers often use anthropomorphic design cues to make robots appear more familiar, approachable, and interaction-capable (Kühne and Peter, 2023). However, anthropomorphism does not

necessarily produce uniformly positive consumer responses, and there is no consensus in the service literature regarding whether customer anthropomorphism facilitates or constrains use intention (Blut et al., 2021). Although human-like cues may increase familiarity and social closeness, they may also change how consumers interpret product information and evaluate the robot's value. Therefore, understanding the conditions under which anthropomorphism improves or weakens consumer attitudes is important for explaining how social robots should be designed and communicated as market-facing products.

To address this question, the present study examines anthropomorphism not as a direct driver of consumer attitudes but as a cue that shapes consumers' perceived psychological distance toward the robot, which in turn alters how different types of product information are interpreted. Specifically, we propose that the persuasive effect of anthropomorphism depends on the congruence between the psychological distance it induces and the type of product information presented to consumers (attribute-focused vs. benefit-focused), and we test this proposition through an online between-subjects experiment.

Theoretical Background

Social Robots and Anthropomorphism

Research on social robots dates back to the early 2000s, when Breazeal (2003) and Fong et al. (2003) laid the conceptual foundation for the field by defining social robots as embodied agents capable of communicating and interacting with humans through social cues, and by distinguishing them from purely functional or industrial robots. Building on this foundation, two decades of research have examined social robots across diverse contexts, including education, healthcare, eldercare, hospitality, and retail (Fong et al., 2003; van Doorn et al., 2017). More recently, the focus of inquiry has shifted from the technical question of how to build socially capable robots to the consumer-behavioral question of how customers perceive, evaluate, and accept robots as service or interaction agents. This shift is reflected in the emergence of marketing-oriented frameworks such as the Automated Social Presence framework (van Doorn et al., 2017) and the Service Robot Acceptance Model (Wirtz et al., 2018), which conceptualize social robots not merely as technologies but as social actors whose acceptance depends on a combination of functional, social-emotional, and relational factors.

Building on this shift, a complementary line of work has examined Human-Robot Interaction (HRI) as a process through which the meaning of social robots is constructed in interaction (Saari et al., 2022; Mahr et al., 2025). Through the social cues they convey, consumers infer the robot's intentions, capabilities, and friendliness, and form relational expectations toward it. In other words, the market acceptance of social robots cannot be explained solely by the range or quality of functions they provide; rather, how the robot is perceived as a social entity plays an important role in shaping consumers' evaluations and acceptance.

Among the design and communication strategies that encourage consumers to perceive social robots as "social entities," anthropomorphism has been one of the most widely used approaches. Anthropomorphism refers to the psychological process of attributing human-like intentions, emotions, and cognitive abilities to nonhuman entities (Epley et al., 2007). In the context of social robots, it has been employed as a strategy to increase accessibility, familiarity, and willingness to interact by making robots appear more "human-like" (Kühne and Peter, 2023). However, there is no consistent consensus regarding the effects of anthropomorphism (Roesler et al., 2021; Li and Suh, 2022; Song et al., 2024). While some studies report that anthropomorphism enhances positive attitudes, trust, and acceptance intentions (Waytz et al., 2014; Moussawi et al., 2021; Yang et al., 2025), others suggest that higher levels of anthropomorphism may evoke discomfort or identity threat (Ferrari et al., 2016; Mende et al., 2019). One classical account of such negative responses is the uncanny valley hypothesis, which posits that as a robot's appearance approaches but does not fully match a human form, observers may experience eeriness or discomfort (Mori et al., 2012). These mixed findings indicate that anthropomorphism may not operate through a single persuasive pathway but may instead elicit different responses depending on contextual conditions and recipient characteristics. Thus, the effects of anthropomorphism are unlikely to translate directly into consumer evaluations and may operate through specific cognitive mediating processes. While prior accounts such as the uncanny valley have often explained responses to human-likeness in terms of affective discomfort or eeriness (Mori et al., 2012), the present study adopts a complementary perspective by examining how anthropomorphism shapes perceived psychological distance and, in turn, alters how consumers respond to different types of product information.

Anthropomorphism and Psychological Distance

To understand why anthropomorphism may shape consumers' perceived psychological distance toward a social robot, we draw on Epley et al.'s (2007) three-factor account of anthropomorphism. According to this account, the tendency to attribute human-like characteristics to nonhuman entities is determined by three motivations, namely elicited agent knowledge, the application of self- or human-relevant knowledge as the inductive basis for reasoning about a novel agent; effectance motivation, the motivation to predict, understand, and act effectively on one's environment; and sociality motivation, the motivation to seek social connection with others (Waytz et al., 2010). Each of these motivations corresponds to a distinct mechanism by which anthropomorphism may bring a social robot closer to the self along the social dimension of psychological distance (Trope and Liberman, 2010).

First, elicited agent knowledge implies that when a nonhuman target carries human-like cues, consumers activate and apply self- or human-relevant schemas to interpret it. In the consumer-behavior literature, schema-based anthropomorphism has been shown to shape how consumers perceive and evaluate products with human-like features (Aggarwal and McGill, 2007; Aggarwal and McGill, 2012). When such schemas are applied, the social robot is interpreted not as an unfamiliar object but as an entity sharing properties with humans, which in turn increases perceived similarity to the self. Perceived similarity is itself a marker of reduced social distance, in that more similar others are construed in more concrete and self-relevant terms (Liviatan et al., 2008). Anthropomorphism may therefore reduce psychological distance by activating a schema that renders the robot more similar to the perceiving consumer.

Second, effectance motivation suggests that anthropomorphism functions to make a nonhuman target more predictable, understandable, and controllable. When consumers face an entity whose behavior is otherwise difficult to interpret, attributing human-like states and intentions provides a familiar knowledge structure that supports inference and anticipation (Epley et al., 2008; Waytz et al., 2010). Empirical evidence in consumer and human-robot interaction contexts indicates that anthropomorphizing a nonhuman target reduces the felt uncertainty associated with it and increases trust in its operation (Waytz et al., 2014). Because reduced uncertainty and increased familiarity are themselves components of reduced social distance (Stephan et al., 2011), anthropomorphism may shorten the perceived gap between the consumer and the social robot by transforming an unfamiliar agent into a more predictable and intelligible one.

Third, sociality motivation refers to consumers' general motivation to form and maintain social connections with others, which can extend to nonhuman targets that display human-like cues. When such cues are present, consumers are more likely to treat the target as a possible partner of social interaction rather than as a mere object, and to form parasocial or relational impressions toward it (Mourey et al., 2017). This tendency may reduce the social distance consumers perceive between themselves and an anthropomorphized social robot. Taken together, the three motivations converge on the prediction that anthropomorphism reduces the psychological distance consumers perceive toward a social robot, and recent evidence in AI assistant contexts has begun to provide direct support for this mediation (Li and Sung, 2021; Blut et al., 2021).

Hypothesis 1. Anthropomorphism reduces the psychological distance that consumers perceive toward a social robot.

Psychological Distance and Persuasive Interpretation of Information

Although anthropomorphism may bring a social robot psychologically closer to the consumer, such changes in psychological distance do not necessarily translate directly into more positive attitudes. According to Construal Level Theory (Trope and Liberman, 2010), psychological distance functions not by directly determining the direction of evaluation but by shaping the cognitive conditions under which the target is interpreted and judged. Specifically, psychologically close targets tend to be represented more concretely, making individuals more sensitive to feasibility-related information, whereas psychologically distant targets tend to be represented more abstractly, leading individuals to place greater weight on high-level desirability-related information such as goals and values (Liberman and Trope, 1998; Amaral and Jiao, 2023). From this perspective, psychological distance is more likely to function as an interpretive framing variable that influences which information is perceived as more diagnostic and persuasive, rather than directly shaping the direction of evaluation (Hernandez et al., 2015).

Within this framing, attribute-focused and benefit-focused appeals occupy distinct positions on the construal-level dimension. According to Hernandez et al. (2015), attributes refer to measurable, concrete, and observable characteristics attached to a product, serving as objective criteria that consumers use to distinguish among alternatives. In contrast, benefits refer to the value or state that consumers ultimately obtain through the use or ownership of a product, emphasizing higher-level outcomes such as happiness, convenience, status, and satisfaction. When consumers perceive a product as psychologically close, attribute-focused appeals emphasizing specific product features or performance may be interpreted as more persuasive, because such information matches the concrete construal evoked by closeness. Conversely, when consumers perceive a product as psychologically distant, benefit-focused appeals emphasizing the product's ultimate value or the changes it brings to one's life may be perceived as more appropriate, because such information matches the abstract construal evoked by distance.

This perspective implies that the effect of anthropomorphism on consumer attitudes toward a social robot operates not directly but through the congruence between anthropomorphism-induced psychological distance and the type of product information consumers encounter. When anthropomorphic cues bring the social robot psychologically closer, attribute-focused appeals are more likely to be interpreted as diagnostic and persuasive. When such cues are absent and the robot is perceived as more distant, benefit-focused appeals are more likely to fit the abstract construal that distance evokes. Although prior research has suggested that anthropomorphism may alter psychological distance (Li and Sung, 2021; Liang et al., 2025), studies that integratively examine how such distance changes translate into persuasive effects through congruence with product information type remain limited.

Hypothesis 2. Product information type moderates the indirect effect of anthropomorphism on attitudes toward social robots through psychological distance.

Hypothesis 2-1. Under attribute-focused appeals, perceiving the social robot as psychologically closer leads to more positive attitudes toward it.

Hypothesis 2-2. Under benefit-focused appeals, perceiving the social robot as psychologically farther leads to more positive attitudes toward it.

Methodology

Research Design and Procedure

Experimental Design

We examined the effects of anthropomorphism and product information type on consumer attitudes toward social robots and analyzed the role of psychological distance in this process. We conducted an online experiment using a 2 (anthropomorphism: high vs. low) × 2 (product information type: attribute-focused vs. benefit-focused) between-subjects factorial design. Participants were randomly assigned to one of the four conditions. To minimize demand characteristics, we introduced the study under the title "User Opinions on Product Advertisements."

Anthropomorphism Manipulation

To manipulate the perceived level of anthropomorphism of the social robot (high vs. low), we presented participants with stimuli and tasks prior to the experimental scenario that combined a visual cue and a cognitive priming task while holding other design features constant. Priming has been widely used in marketing and consumer behavior research as a method for temporarily directing individuals' thoughts and goals in a specific direction required by the researcher (Herr, 1986; Dijksterhuis and Van Knippenberg, 1998; Minton et al., 2017). The visual stimuli were based on a single robot model, shown in Figure 1. We deliberately held the robot's body shape, size, color, and functional structure constant across the two conditions to avoid uncontrolled variation that could confound the manipulation. In the high-anthropomorphism condition, the front of the robot included morphological elements corresponding to eyes and a mouth, creating an impression similar to a human face. In the low-anthropomorphism condition, the facial elements were removed while the functional structure was maintained. The face is one of the most

critical social cues in human perception (Aggarwal and McGill, 2007), and face-like cues are known to activate human schemas and increase anthropomorphic perceptions (Landwehr et al., 2011; Ren et al., 2024).

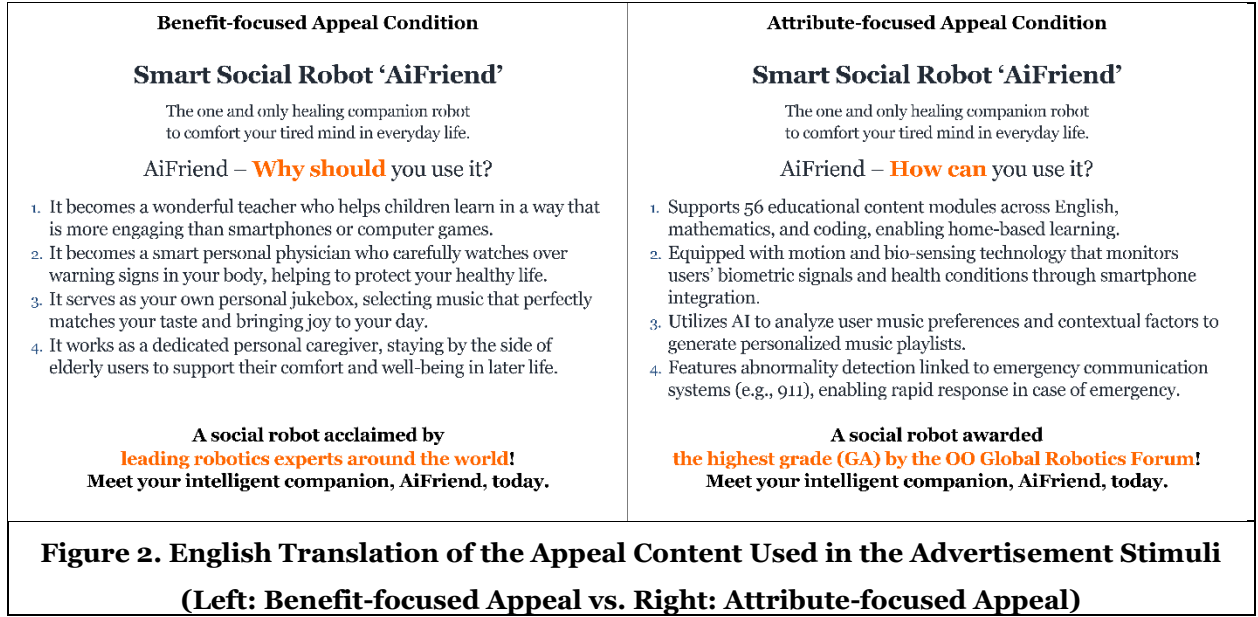


Figure 1. Stimuli by Level of Anthropomorphism (Left: Low vs. Right: High)

The visual cue was reinforced by a condition-specific cognitive priming task that operates on a complementary process. While visual cues serve as externally provided morphological inputs, the cognitive task engages participants in actively projecting human-like attributes onto the target (Eskine and Locander, 2014; Chandler and Schwarz, 2010; Waytz et al., 2014). In the high-anthropomorphism condition, participants were asked to name the robot and describe its personality and traits (“Please give this product a suitable name,” “What kind of personality does this product have?”, “What kind of traits does this product possess?”). In the low-anthropomorphism condition, participants were asked to describe the robot in terms of functional and categorical attributes (“Which product category do you think this product belongs to?”, “What functions do you think this product includes?”, “What material does this product appear to be made of?”).

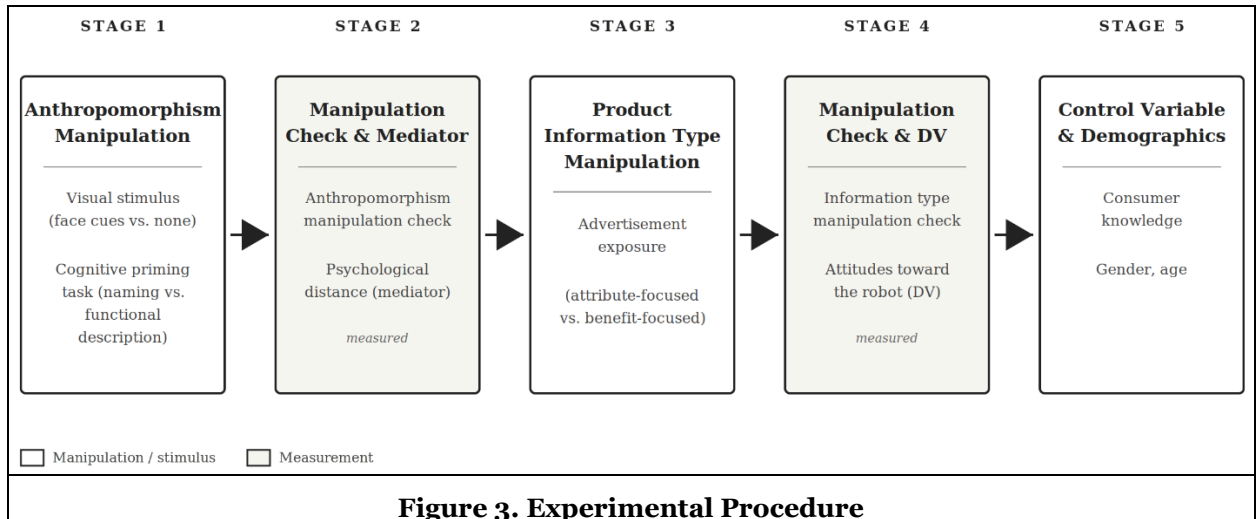
Product Information Type Manipulation

We manipulated product information type by varying the focus of the advertising message while keeping the social robot constant across conditions. In the attribute-focused condition, the advertisement emphasized measurable and objective attributes such as quantitative figures, product functions, and performance specifications. Evaluation information was attributed to clearly identifiable and verifiable sources such as official ratings or certifications from professional institutions, strengthening the concreteness and diagnosticity of the message. In the benefit-focused condition, the advertisement emphasized abstract and outcome-oriented benefits such as the ultimate value obtained through product use, emotional satisfaction, and changes in everyday life. Evaluation information was presented in a generalized and symbolic form of recognition, such as statements that the product had received positive responses and high approval overall, reinforcing the abstract and outcome-oriented interpretation of the message. We designed the stimuli in both conditions to maintain the same length and structural elements while differing only in message focus. The original stimuli were created and presented to participants in Korean. To improve readability, Figure 2 presents an English translation of the appeal content used in the two advertisement stimuli, rather than the full original stimulus layout.



Procedure

The procedure followed five stages, as summarized in Figure 3. Participants first completed the anthropomorphism manipulation (Stage 1), then responded to manipulation check items for anthropomorphism and to the psychological distance measure (Stage 2). They were then exposed to the advertisement stimulus corresponding to the assigned product information type condition (Stage 3), after which they responded to manipulation check items for product information type and to the attitudes measure (Stage 4). Finally, participants responded to the consumer knowledge measure used as a control variable, followed by demographic items (Stage 5). We placed each manipulation check immediately after the corresponding manipulation but before the next stimulus, so that downstream stimuli could not retroactively influence participants’ perceptions of a prior manipulation. After completing all measures, participants received a small participation incentive in accordance with the operating standards of the survey company.



Participants

We recruited participants in their 20s to 40s from an online panel managed by a renowned survey company in South Korea. Participants accessed the online survey page and participated in the experiment through a link included in an email invitation sent by the survey company. Our study received approval from the Institutional Review Board (IRB) of the authors' affiliated institution. All participants were provided with information regarding the research purpose, procedures, and privacy protection policy, and only those who explicitly indicated their voluntary consent were allowed to participate in the survey. A consent form for research participation was presented on the initial screen of the survey, and participants were allowed to proceed to the next stage only after indicating their agreement.

Prior to data collection, we conducted an a priori power analysis using G*Power 3.1 to determine the minimum sample size required for the analysis. Assuming an effect size of $f^2 = .05$, a significance level of $\alpha = .05$, and statistical power of $1 - \beta = .80$, the required minimum sample size for a regression model including an interaction effect and seven predictors was calculated to be 159. Accordingly, we recruited participants to meet this requirement.

To ensure comparability across conditions and internal validity, participants were recruited and assigned so that they were evenly distributed across the four experimental conditions. In addition, to avoid bias toward specific demographic groups, sampling was conducted to achieve a relatively balanced distribution across gender and age groups (20s, 30s, and 40s). This procedure was intended to minimize potential confounding effects resulting from demographic differences across conditions. As a result, a total of 206 responses were used for the final analysis after excluding incomplete surveys and inattentive responses.

The number of participants in each condition was distributed as follows. In the attribute-focused condition, 52 participants were assigned to the low-anthropomorphism condition and 51 to the high-anthropomorphism condition (total $n = 103$). In the benefit-focused condition, 52 participants were assigned to the low-anthropomorphism condition and 51 to the high-anthropomorphism condition (total $n = 103$). Overall, the sample consisted of 104 participants in the low-anthropomorphism condition and 102 in the high-anthropomorphism condition. The gender distribution of respondents was 49% male (101 participants) and 51% female (105 participants). The age distribution was relatively balanced, with 29.1% in their 20s (60 participants), 34.0% in their 30s (70 participants), and 36.9% in their 40s (76 participants). All responses were collected anonymously, and the collected data were used exclusively for research purposes.

Measurement Items

We measured anthropomorphism as the independent variable, psychological distance as the mediator, and attitudes toward social robots as the dependent variable. In addition, we included manipulation check items to verify the appropriateness of the product information type manipulation. We also measured consumer knowledge, gender, and age as control variables, considering that respondents' product-related knowledge and demographic characteristics could influence the formation of attitudes toward social robots. Measurement items for each construct were developed based on conceptual definitions and scales used in relevant prior studies and were adapted to fit the purpose and context of our study.

Anthropomorphism refers to the extent to which the presented product is perceived as similar to a human. We measured anthropomorphism using two items based on Kim and McGill (2011), Kim et al. (2016), and Chen et al. (2017). Specifically, participants responded to the following items on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree): "The product in the picture looks like a person," and "The product in the picture feels like a person" (Cronbach's $\alpha = .918$).

To measure psychological distance, we developed items based on the conceptual definition proposed by Trope and Liberman (2010), focusing on how close the presented robot was perceived to be by the participant. Participants responded to four items on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree): "I feel psychologically close to the product in the picture," "The product in the picture feels like a close friend," "I think I could form a close relationship with the product in the picture," and "I think the product in the picture could become close to me" (Cronbach's $\alpha = .938$).

For the manipulation check of product information type, we developed items based on Escalas and Luce (2003), Trope and Liberman (2010), and Hernandez et al. (2015). Participants evaluated the extent to

which the advertisement emphasized product attributes or benefits by responding to four items using a 7-point semantic differential scale. The items were: “This advertisement explains how/why the product should be used,” “The message delivered by this advertisement is concrete/abstract,” “The message delivered by this advertisement is process-focused/outcome-focused,” and “The message delivered by this advertisement focuses more on product attributes/the benefits obtained from using the product” (Cronbach’s $\alpha = .85$).

Attitudes toward the social robot, the dependent variable, were measured using four items adapted from Mitchell and Olson (1981) and Batra and Ray (1986). The items were: “I like the product introduced in this advertisement,” “I find the product introduced in this advertisement appealing,” “I feel favorable toward the product introduced in this advertisement,” and “I would like to purchase the product introduced in this advertisement.” Participants responded to these items using a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree) (Cronbach’s $\alpha = .95$).

We included consumer knowledge about social robots, gender, and age as control variables. Consumer knowledge refers to “the amount of information and level of understanding that consumers possess about a particular product category” (Flynn and Goldsmith, 1999). Consumers with greater knowledge about a product generally tend to interpret message content more elaborately and process more information than those with less knowledge. Consumer knowledge is also related to the level of attribute information used in product evaluation. Consumers with higher knowledge levels tend to process information based on product attributes, whereas those with lower knowledge levels tend to focus more on the outcomes or benefits of product use (Maheswaran and Sternthal, 1990). Accordingly, we controlled for consumer knowledge because prior knowledge about social robots could influence how participants interpreted attribute-focused and benefit-focused product information. Specifically, we measured subjective knowledge using two items on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree): “Compared with other users, I know a lot about social robots,” and “I have a considerable amount of information about social robot products” (Cronbach’s $\alpha = .92$).

Gender may shape social robot evaluation because men and women may rely on different evaluation criteria. Prior research shows that women tend to place greater weight on social influence and perceived enjoyment when accepting social robots. In contrast, men tend to place greater weight on perceived usefulness and ease of use (Forgas-Coll et al., 2022). Thus, we controlled for gender because the weight placed on social, experiential, functional, and usability-related criteria may shape responses to benefit-focused and attribute-focused product information.

Age may also shape social robot evaluation because social robots are technology-based products. A meta-analysis on technology acceptance shows that age is related to specific technology perceptions, particularly perceived ease of use. It also shows that these relationships depend on technology type (Hauk et al., 2018). Recent advertising research also reports age-group differences in cognitive evaluative measures of advertising effectiveness, such as associative penetration, purchase-funnel patterns, and product category knowledge (Mecredy et al., 2025). Therefore, we controlled for age to account for possible age-related differences in technology evaluation and advertising response.

Research Results

We used SPSS 29.0 to test our hypotheses through three analyses. First, a one-way ANCOVA tested the effect of anthropomorphism on psychological distance. Second, a two-way ANCOVA tested the interaction effect of anthropomorphism and product information type on attitudes toward social robots. Third, a moderated mediation analysis using Model 14 of the PROCESS macro (version 5.0; Hayes, 2022) examined whether the indirect effect of anthropomorphism on attitudes through psychological distance varies by product information type. Gender, age, and consumer knowledge were included as control variables in all analyses, and the significance of indirect effects was tested using bootstrapping with 5,000 resamples.

Manipulation Check

We conducted manipulation checks to verify whether participants perceived the manipulations in the intended direction. Participants in the high-anthropomorphism condition evaluated the presented robot as significantly more human-like than those in the low-anthropomorphism condition ($M_{\text{AntHigh}} = 4.88$ vs.

$M_{\text{AntLow}} = 3.86$, $F(1, 204) = 30.82$, Partial $\eta^2 = .13$, $p < .001$). A cross-check on whether the anthropomorphism manipulation affected perceptions of product information type revealed no significant difference ($M_{\text{AntHigh}} = 4.26$ vs. $M_{\text{AntLow}} = 4.08$, $F(1, 204) = .94$, Partial $\eta^2 = .01$, ns). The product information type manipulation check scores differed significantly between the two conditions ($M_{\text{Benefit}} = 4.92$ vs. $M_{\text{Attribute}} = 3.43$, $F(1, 204) = 93.09$, Partial $\eta^2 = .31$, $p < .001$). A cross-check showed that the product information type manipulation did not significantly affect anthropomorphism perception ($M_{\text{Benefit}} = 4.48$ vs. $M_{\text{Attribute}} = 4.26$, $F(1, 204) = 1.23$, Partial $\eta^2 = .01$, ns).

Hypothesis Test

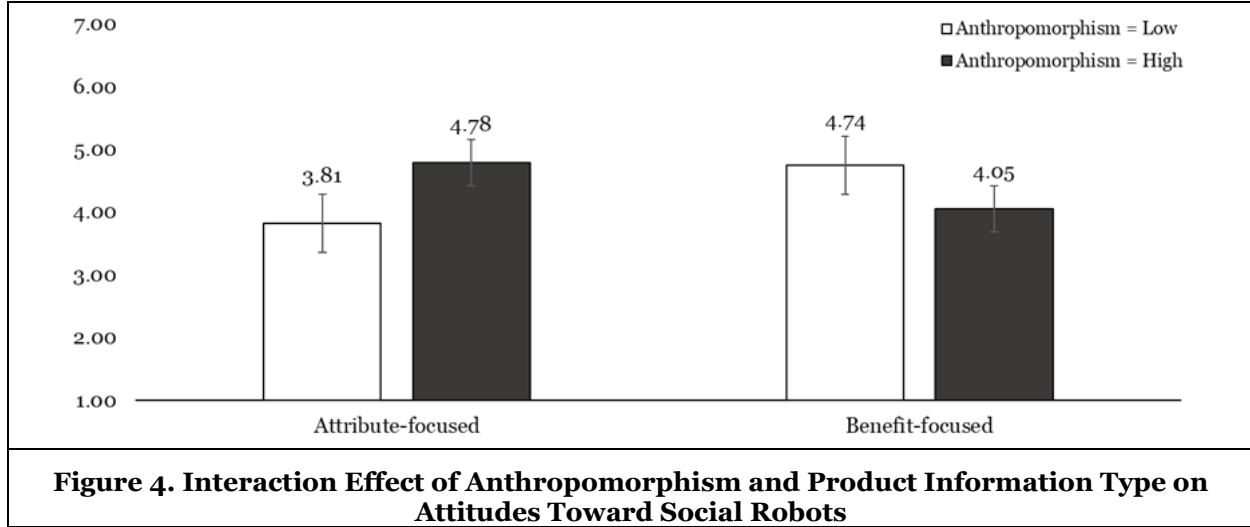
To examine the effect of anthropomorphism level on consumers' perceived psychological distance, we conducted an ANCOVA with consumer knowledge, gender, and age as control variables. Higher mean values indicate that the social robot is perceived as psychologically closer. The results show that participants perceived the social robot as psychologically closer in the high-anthropomorphism condition than in the low-anthropomorphism condition ($M_{\text{AntHigh}} = 4.91$ vs. $M_{\text{AntLow}} = 3.93$, $F(1, 201) = 31.34$, Partial $\eta^2 = .14$, $p < .001$). Regarding the covariates, gender, age, and consumer knowledge did not have significant effects on psychological distance.

Table 1. ANCOVA Results for Psychological Distance			
Source of Variation	$F(df1, df2)$	p	Partial η^2
Gender	.02 (1, 201)	.90	.00
Age	1.37 (1, 201)	.24	.01
Consumer Knowledge	.72 (1, 201)	.40	.00
Anthropomorphism	31.34 (1, 201)	< .001	.14
Note: Dependent variable = Psychological Distance. Covariates include gender, age, and consumer knowledge.			

Next, to examine the effects of anthropomorphism level and product information type on attitudes toward social robots, we conducted a two-way ANCOVA including consumer knowledge, gender, and age as control variables. The results revealed a statistically significant interaction effect between anthropomorphism level and product information type ($F(1, 199) = 21.72$, Partial $\eta^2 = .10$, $p < .001$). Regarding the covariates, gender and age did not have significant effects on attitudes toward social robots. Consumer knowledge had a significant effect ($F(1, 199) = 6.37$, Partial $\eta^2 = .03$, $p < .05$). These results indicate that participants' prior knowledge about social robots was related to their attitudes toward social robots. Importantly, the interaction effect between anthropomorphism and product information type remained significant after controlling for gender, age, and consumer knowledge.

Table 2. Two-Way ANCOVA Results for Attitudes toward Social Robots			
Source of Variation	$F(df1, df2)$	p	Partial η^2
Gender	1.10 (1, 199)	.30	.01
Age	3.87 (1, 199)	.05	.02
Consumer Knowledge	6.37 (1, 199)	.01	.03
Anthropomorphism (A)	.59 (1, 199)	.44	.00
Product Information Type (B)	.36 (1, 199)	.55	.00
A $\times$ B	21.72 (1, 199)	< .001	.10
Note: Dependent variable = Attitudes toward Social Robots. Covariates include gender, age, and consumer knowledge.			

To examine the interaction effect in greater detail, we analyzed the simple main effects of anthropomorphism within each product information type condition. The results showed that under the attribute-focused appeal condition, participants reported more positive attitudes in the high-anthropomorphism condition than in the low-anthropomorphism condition ($M_{\text{AntHigh}} = 4.78$ vs. $M_{\text{AntLow}} = 3.81$, $F(1, 98) = 14.11$, Partial $\eta^2 = .13$, $p < .001$). In contrast, under the benefit-focused appeal condition, the mean attitude in the high-anthropomorphism condition was lower than that in the low-anthropomorphism condition ($M_{\text{AntHigh}} = 4.05$ vs. $M_{\text{AntLow}} = 4.74$, $F(1, 98) = 7.67$, Partial $\eta^2 = .07$, $p = .007$). These results indicate that the interaction between anthropomorphism and product information type has a significant effect on attitudes toward social robots. Specifically, anthropomorphism facilitates the formation of positive attitudes under the attribute-focused appeal condition, whereas the effect does not emerge in the same direction under the benefit-focused appeal condition.



Conditions		Mean	SD	N
Product Information Type	Anthropomorphism			
Attribute-focused Appeal	High-anthropomorphism	4.78	1.13	51
	Low-anthropomorphism	3.81	1.37	52
Benefit-focused Appeal	High-anthropomorphism	4.05	1.44	51
	Low-anthropomorphism	4.74	.98	52

To examine the psychological mechanism underlying the interaction effect, we conducted the moderated mediation analysis using the PROCESS macro (Model 14). The mediator, psychological distance, was mean-centered to facilitate the interpretation of the interaction term. The results show that in the mediator model, anthropomorphism had a significant positive effect on psychological distance ($B = 0.92$, $SE = 0.02$, $t = 51.65$, $p < .001$). This result indicates that when a social robot is anthropomorphized, consumers tend to perceive the robot as psychologically closer. Next, in the model in which attitudes toward social robots were specified as the dependent variable, the interaction effect between psychological distance and product information type was statistically significant ($B = -1.11$, $SE = 0.11$, $t = -10.19$, $p < .001$). This result suggests that the effect of psychological distance on attitudes toward social robots varies depending on the type of product information presented. In contrast, the direct effect of anthropomorphism was not statistically significant ($B = -0.22$, $SE = 0.19$, $t = -1.14$, ns). Regarding the covariates in the moderated mediation model, consumer knowledge ($B = -0.00$, $SE = 0.02$, $t = -0.22$, ns), gender ($B = 0.01$, $SE = 0.05$, $t = 0.28$, ns), and age ($B = 0.00$, $SE = 0.00$, $t = 0.88$, ns) did not significantly predict psychological distance. In the attitude model, consumer knowledge had a significant positive effect on attitudes toward social robots ($B = 0.15$, $SE = 0.05$,

$t = 2.95, p < .01$). However, gender ($B = 0.18, SE = 0.14, t = 1.26, ns$) and age ($B = 0.01, SE = 0.01, t = 0.59, ns$) did not have significant effects.

Table 4. Results of Moderated Mediation Analysis (PROCESS Model 14)						
Dependent Variable	Predictor	B	SE	t	p	95% CI
PD (M)	Constant	-4.10	0.17	-24.10	< .001	[-4.44, -3.77]
	CK	-0.00	0.02	-0.22	0.82	[-.04, .03]
	Gender	0.01	0.05	0.28	0.78	[-.08, .11]
	Age	0.00	0.00	0.88	0.38	[-.00, .01]
	ANT (X)	0.92	0.02	51.65	< .001	[.88, .95]
	Model Statistics	R ² = .93*, F(4,201) = 674.12, p < .001				
ATT (Y)	Constant	4.33	0.97	4.48	< .001	[2.42, 6.23]
	CK	0.15	0.05	2.95	< .01	[.05, .26]
	Gender	0.18	0.14	1.26	0.21	[-.10, .46]
	Age	0.01	0.01	0.59	0.55	[-.01, .02]
	ANT (X)	-0.22	0.19	-1.14	0.25	[-.60, .16]
	PD (M)	1.00	0.21	4.73	< .001	[.58, 1.42]
	PIT (W)	0.06	0.14	0.45	0.66	[-.22, .35]
	PD × PIT	-1.11	0.11	-10.2	< .001	[-1.32, -.89]
	Model Statistics	R ² = .41, F(7,198) = 19.33, p < .001				
Note: Gender: 1 = Male, 2 = Female; ANT = Anthropomorphism (X), PD = Psychological Distance (M), PIT = Product Information Type (W), CK = Consumer Knowledge (covariate). Gender, age, and consumer knowledge were included as covariates. Coefficients are rounded to two decimal places, and values smaller than .005 are reported as ±.00. * The high R ² of the mediator model does not indicate a discriminant validity problem between anthropomorphism and psychological distance. The HTMT ratio between the two constructs was .669, well below the conservative threshold of .85 (Henseler et al., 2015), supporting discriminant validity.						

We further examined the conditional effect of psychological distance according to product information type. The results showed that under the attribute-focused appeal condition, attitudes toward social robots became significantly more positive as psychological distance became closer ($B = 1.00, SE = 0.21, t = 4.73, p < .001$). In contrast, under the benefit-focused appeal condition, the effect of psychological distance on attitudes toward social robots was not statistically significant ($B = -0.10, SE = 0.21, t = -0.49, p = .62$).

Table 5. Conditional effects of M on Y at values of W					
PIT (W)	Effect	SE	t	p	95% CI
Attribute-focused (0)	1.00	.21	4.73	< .001	[.58, 1.42]
Benefit-focused (1)	-.10	.21	-.49	.62	[-.51, .31]

Finally, we tested the conditional indirect effect of anthropomorphism on attitudes toward social robots through psychological distance using a bootstrapping procedure. The results showed that under the attribute-focused appeal condition, the indirect effect of anthropomorphism on attitudes toward social robots through psychological distance was significant (Indirect effect = 0.92, Boot SE = 0.18, 95% CI [0.57, 1.26]). In contrast, under the benefit-focused appeal condition, this indirect effect was not statistically significant (Indirect effect = -0.09, Boot SE = 0.20, 95% CI [-0.48, 0.30]). In addition, the index of moderated mediation was significant (Index = -1.01, Boot SE = 0.14, 95% CI [-1.25, -0.72]), indicating that the indirect effect of anthropomorphism on attitudes toward social robots through psychological distance significantly differs depending on the type of product information.

Table 6. Conditional indirect effects of X on Y through M at values of W			
PIT (W)	Effect	Boot SE	95% Boot CI
Attribute-focused (0)	0.92	0.18	[0.57, 1.26]
Benefit-focused (1)	-0.09	0.20	[-0.48, 0.30]

Discussion

We examined the effect of anthropomorphism in social robots on consumers' attitudes from the perspective of psychological distance and examined how this relationship varies by product information type. The results show that anthropomorphized social robots are perceived as psychologically closer to consumers than non-anthropomorphized robots. In addition, the effect of psychological distance on attitudes toward social robots varies depending on the type of product information. Specifically, under the attribute-focused appeal condition, attitudes toward social robots became more positive as psychological distance decreased, and the indirect effect of anthropomorphism on attitudes through psychological distance was also significant. In contrast, under the benefit-focused appeal condition, neither the conditional effect of psychological distance nor the indirect effect was statistically significant. Taken together, Hypothesis 1 and Hypothesis 2-1 were supported, whereas Hypothesis 2-2 was not.

Our findings suggest that anthropomorphism of social robots may be associated with consumers' perceptions of psychological distance in the cognitive evaluation process. Anthropomorphism can lead individuals to perceive nonhuman targets as human-like entities, thereby reducing the psychological gap between the target and the self (Epley et al., 2007; Jiang et al., 2024) and encouraging consumers to view the target as a more social and interaction-capable entity. Consistent with this perspective, the results show that anthropomorphized social robots can be perceived as psychologically closer targets by consumers.

We also show that the effect of psychological distance varies depending on its congruence with the type of product information. According to Construal Level Theory, psychologically close targets elicit greater sensitivity to concrete and feasibility-focused information (Trope and Liberman, 2010). Because attribute-focused information emphasizes specific product characteristics such as functions or performance, consumers who perceive the target as psychologically close may form more positive evaluations when exposed to such information. Consistent with this explanation, attitudes toward social robots became more positive as psychological distance decreased under the attribute-focused appeal condition.

Interestingly, under the benefit-focused appeal condition, the conditional effect of psychological distance was not statistically significant. Although this mechanism was not directly tested in our study, this result suggests that anthropomorphism may exert additional influences on consumers' information interpretation processes in the context of social robots. Prior research indicates that anthropomorphized targets tend to be perceived not as mere objects but as agents with intentions or purposes (Epley et al., 2007; Waytz et al., 2010), and such perceptions may lead consumers to interpret the target as a subject of social interaction.

One possible explanation is that anthropomorphism may involve more than psychological distance alone. According to Construal Level Theory, closer psychological distance is generally associated with more concrete construal, whereas greater psychological distance is associated with more abstract construal (Trope and Liberman, 2010). This logic is consistent with our finding that attribute-focused information was more strongly linked to attitudes when anthropomorphic cues reduced psychological distance. However, benefit-focused appeals may require consumers to construe broader outcomes or roles of the product, and such interpretation may depend on additional perceptions not measured in this study. Prior research suggests that anthropomorphism can be associated with perceptions of mind or agency (Epley et al., 2007; Waytz et al., 2010), but the present study did not directly examine whether such perceptions were activated or whether they shaped message interpretation. Thus, the non-significant indirect effect under benefit-focused appeals should be interpreted cautiously. Future research should test this possibility by jointly measuring psychological distance and mind-related constructs such as perceived agency, intention inference, and perceived interaction capability.

Conclusion

We examined the effect of anthropomorphism of social robots on consumer attitudes through the cognitive mechanism of psychological distance and investigated how this relationship varies depending on product information type. To do so, we conducted an experiment manipulating anthropomorphism level and product information type. The results show that anthropomorphized social robots are perceived as psychologically closer to consumers than non-anthropomorphized robots. However, anthropomorphism did not directly influence attitudes toward social robots; instead, its effect emerged indirectly through psychological distance. Furthermore, this indirect effect varied depending on product information type. Specifically, under the attribute-focused appeal condition, attitudes toward social robots became more positive as psychological distance became closer, and anthropomorphism also exerted a significant indirect effect on attitudes through psychological distance. In contrast, under the benefit-focused appeal condition, neither the conditional effect of psychological distance nor the indirect effect of anthropomorphism was statistically significant. Our findings indicate that anthropomorphism does not produce a uniformly positive effect on consumer evaluations but rather operates conditionally when combined with specific informational contexts.

We make several theoretical contributions. First, we empirically identify the cognitive mechanism through which anthropomorphism influences consumer responses in the context of HRI. Prior research has often described anthropomorphism as a factor that directly influences consumers' attitudes. However, our results suggest that anthropomorphism may operate indirectly by altering perceptions of psychological distance rather than directly influencing attitudes. These findings complement prior approaches that explain anthropomorphism effects primarily through relational or emotional responses and highlight the importance of understanding anthropomorphism from the perspective of cognitive processes through which consumers interpret technological products such as robots.

Second, we extend the application of psychological distance research to the context of intelligent technological products such as social robots. Previous research based on Construal Level Theory has primarily shown that psychological distance influences which types of information consumers interpret as more diagnostic. Our findings confirm that this information interpretation mechanism can also emerge in evaluations of social robots, while also demonstrating that anthropomorphism can function as an antecedent that alters perceptions of psychological distance and thereby influences this interpretive process.

Third, we identify a boundary condition under which anthropomorphism effects operate. Prior research has reported conflicting findings regarding whether anthropomorphism has positive or negative effects on consumer responses. Our results suggest that these differences may arise not because the effect of anthropomorphism itself is unstable, but because the effect varies depending on the congruence between the psychological distance created by anthropomorphism and the type of product information presented. These findings imply that anthropomorphism effects may depend on the context of product communication and the way information is presented.

Our study also provides several practical contributions. First, we suggest a tentative pattern that may inform future communication design for social robots. The results indicate that when the level of anthropomorphism is increased, such as by incorporating human-like elements such as eyes or facial expressions into the robot's appearance, attribute-focused information appears more compatible with the closer psychological distance such cues induce. This finding is most relevant to early-stage message development, where managers and designers explore how robot appearance and communication framing may be combined. Specifically, when a robot is designed with prominent human-like cues, our findings suggest that attribute-focused messages that highlight the robot's concrete features, functions, and performance are more likely to be received favorably than benefit-focused messages that emphasize abstract outcomes.

Second, we show that anthropomorphic design in social robot marketing does not always strengthen persuasive effects. The results indicate that under benefit-focused messages emphasizing the outcomes or advantages obtained from using the robot, the effect of psychological distance was not significant. This finding suggests that anthropomorphic design may not produce the same persuasive effect across all types of marketing messages in the promotion of social robots. Therefore, firms should design marketing strategies by considering both the type of advertising message and the communication context rather than relying solely on anthropomorphic design.

Third, our findings point to the potential value of considering anthropomorphic design and marketing communication jointly rather than as independent decisions. Our results indicate that anthropomorphic design does not independently determine consumer evaluations; rather, its effect depends on its congruence with the type of advertising message. This pattern suggests that the alignment between design cues and message framing is a useful consideration alongside other product development priorities. In particular, when product teams iterate on anthropomorphic design choices, our findings indicate that the choice of message framing in accompanying communication is not independent of those design choices and merits attention as part of the same design-and-communication workflow rather than as a separate downstream decision.

Although we provide theoretical and practical contributions by identifying the relationships among anthropomorphism, psychological distance, and advertising message type, several limitations remain. First, the generalizability of our findings warrants further investigation with respect to stimulus diversity, the scope of anthropomorphism cues, and cultural scope. The current study relied on a single social robot stimulus, and our anthropomorphism manipulation combined facial cues with a cognitive priming task while holding other design features constant. As a result, our findings should be interpreted as effects observed within this particular controlled manipulation rather than as effects that necessarily generalize to all forms of anthropomorphized robots. Whether the identified effects extend to robots that vary in levels of physical anthropomorphism, functional features, or design characteristics remains to be examined, and it likewise remains open whether the effects hold under anthropomorphism manipulations that involve non-visual cues such as voice, movement, or interaction style, or that draw on visual cues other than facial features, such as body shape, size, or proportions. Similarly, the sample was limited to South Korean consumers in their 20s to 40s. Given prior evidence that robot acceptance, design preferences, and responses to robot appearance differ across cultural contexts (Li et al., 2010), the cross-cultural generalizability of our findings should be interpreted with caution. Replicating and extending our findings across a broader range of robot stimuli, a wider set of anthropomorphism cues, and culturally diverse samples would be a promising avenue for establishing the boundary conditions and cross-cultural robustness of the effects we report.

Second, because we employed an experimental design based on static advertising stimuli, our findings may not fully capture consumer responses that emerge in actual HRI contexts. Future research should examine whether anthropomorphism effects appear similarly in real usage contexts by employing research designs involving actual interactions with robots, video-based stimuli, or field experiments.

Third, while psychological distance accounted for the anthropomorphism effect under attribute-focused appeals, it did not do so under benefit-focused appeals. As discussed above, this pattern may indicate that anthropomorphism involves more than psychological distance alone, and that benefit-focused interpretation may depend on additional perceptions, such as those related to mind or agency, that were not measured in the present study. Because we did not assess these mind-related constructs, this possibility remains untested in our data. Future research that jointly measures psychological distance and mind-related constructs such as perceived agency, intention inference, and perceived interaction capability would allow a direct test of this possibility and help clarify the conditions under which psychological distance functions as the operative mechanism linking anthropomorphism to consumer attitudes.

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